

Cover Letter

August 10, 2026

To the Editor-in-Chief, *Computers & Security*

Dear Editor-in-Chief,

We submit our manuscript entitled "**Evidence-Preserving Graph Denoising for Noisy-Label Intrusion Detection: A Leakage-Audited Study of When Graph Structure Helps**" for consideration as a research article in *Computers & Security*.

Scope. Intrusion-detection classifiers are routinely trained on delayed, rule-generated, or inconsistent labels. A common remedy—deleting suspicious samples—is risky in security operations because the deleted instances are often the rare attack samples analysts most need. We revisit the CoLD collaborative label-denoising framework and ask two questions: does a multi-view, label-space graph consistency diagnostic (Graph-CDM) improve robustness, and does evidence-preserving soft weighting protect rare-attack evidence better than hard deletion?

Contributions. The study is, deliberately, an audit before it is a method:

1. A label-space multi-view graph diagnostic that modestly but significantly

improves aggregate Macro-F1 over its aligned CoLD baseline on three real datasets (CICIDS-2017, CESNET-TLS-Year22, UNSW-NB15) under a leakage-audited protocol.

2. A mechanism-level finding: evidence-preserving soft weighting is

statistically indistinguishable from hard deletion on aggregate accuracy, yet significantly improves rare/tail attack-class Macro-F1 under high asymmetric noise at no aggregate cost—within the shared pipeline.

3. An honest external baseline ordering under the identical protocol:

Confident Learning outperforms our method on all datasets, while full Co-Teaching collapses on rare classes—empirically confirming that deletion- and forgetting-based mechanisms can sacrifice rare-attack evidence unless the noise diagnostic is precise.

4. A leakage audit showing that graph denoising results in security are easy

to inflate (24% exact duplicates in CICIDS-2017; clean-label signal paths), collapsing an apparent +28-point advantage to a defensible +6.7 points, together with a graph-informativeness diagnostic bounding when graph structure helps.

We believe the audit discipline and the audited baseline ordering are of direct practical relevance to the journal's readership, independent of the specific method.

Statements. This manuscript is original, has not been published before, and is not under consideration by any other journal. All authors have approved the submission. Raw datasets are public; code, frozen result hashes, and the reproduction command sequence are available as described in the Data Availability section.

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Sincerely, Tianqi Guo (on behalf of all authors)